

Table 68. WLCSP19 - Mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A ⁽²⁾	-	-	0.577	-	-	0.0227
A1 ⁽³⁾	0.152	-	-	0.0060	-	-
A2	-	0.380	-	-	0.0150	-
A3	-	0.025	-	-	0.0010	-
b ⁽⁴⁾	0.215	0.240	0.265	0.0085	0.0094	0.0104
D ⁽⁵⁾	2.492 BSC			0.0981 BSC		
D1 ⁽⁵⁾	1.750 BSC			0.0689 BSC		
E ⁽⁵⁾	1.643 BSC			0.0647 BSC		
E1 ⁽⁵⁾	0.909 BSC			0.0358 BSC		
eD ⁽⁵⁾⁽⁶⁾	0.350 BSC			0.0138 BSC		
eS ⁽⁵⁾⁽⁶⁾	0.350 BSC			0.0138 BSC		
N ⁽⁷⁾	19					
SD ⁽⁵⁾⁽⁸⁾	0.175 BSC			0.0069 BSC		
SE ⁽⁵⁾⁽⁸⁾	0.152 BSC			0.0060 BSC		
aaa ⁽⁹⁾	0.100			0.0039		
bbb ⁽⁹⁾	0.100			0.0039		
ccc ⁽⁹⁾	0.100			0.0039		
ddd ⁽⁹⁾	0.050			0.0020		
eee ⁽⁹⁾	0.050			0.0020		

1. Values in inches are converted from mm and rounded to 4 decimal digits.
2. The profile height A is the distance from the seating plane to the highest point on the package. It is measured perpendicular to the seating plane.
3. A1 is defined as the distance from the seating plane to the lowest point on the package body.
4. Dimension b is measured at the maximum diameter of the terminal (ball) in a plane parallel to Datum C.
5. BSC stands for BASIC dimensions. It corresponds to the nominal value and has no tolerance. For tolerances, refer to form and position table. On the drawing, these dimensions are framed. For the tolerances, refer to form and position values.
6. e represents the solder balls grid pitch(es).
7. N represents the total number of balls.
8. Basic dimensions SD & SE are defining the ball matrix position with respect to datums A and B.
9. Tolerance of form and position drawing.